SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2014

Paper 1

H1 BIOLOGY
8875

Friday
29 August 2014

1 hour

Additional materials:
Optical Mark Sheet

INSTRUCTIONS TO CANDIDATES

Write you name, CG and index number in the spaces at the top of this page.

Enter your name, subject title, test name, class. For your index number, enter your full NRIC number. Shade the corresponding lozenges on the OMS according to the instructions given by the invigilators.

There are forty questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C**, **D**. Choose the **one** you consider correct and record your choice in **soft pencil** on the OMS.

AT THE END OF THE EXAMINATION, HAND IN BOTH OMS AND QUESTION PAPERS.

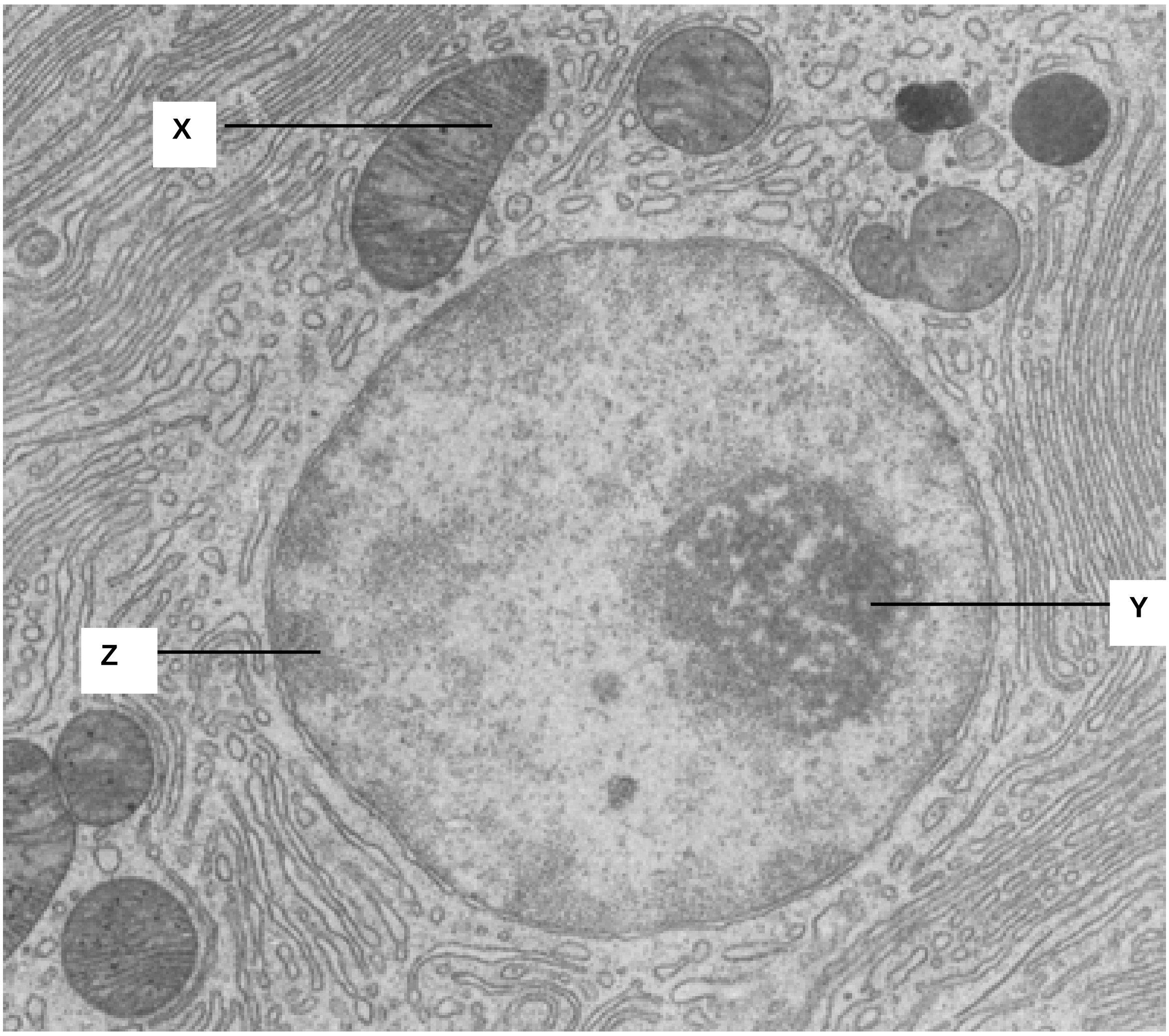
INFORMATION FOR CANDIDATES

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done on the question paper.

This question paper consists of 24 printed pages.

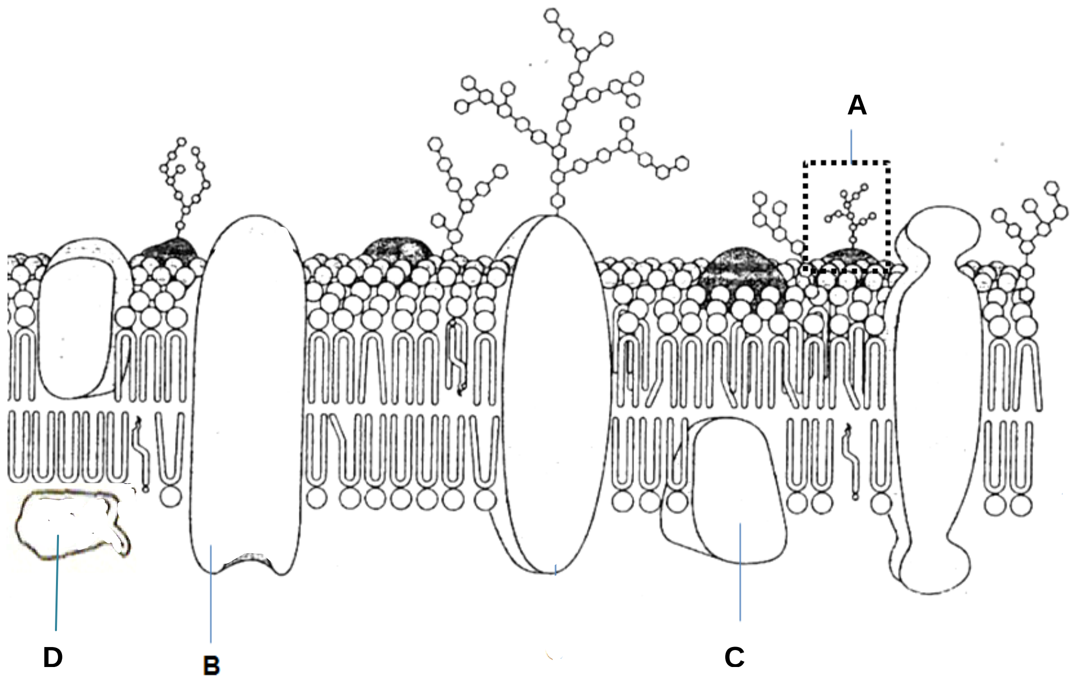
1. The electron micrograph below shows the structures found in a cell.



Which of the following statements is true for structures **X**, **Y** and **Z**?

	X	Y	Z
A	Contains reduced NAD^+ and reduced FAD	Transcription of gene coding for ribosomal RNA	Contains heterochromatin which is transcriptionally inactive
B	Involved in oxidative phosphorylation	Transcription of gene coding for ribosomal protein	Contains heterochromatin which is transcriptionally active
C	Contains reduced NAD^+ and reduced FAD	Involved in assembly of ribosomal subunits	Contains euchromatin which is transcriptionally inactive
D	Involved in oxidative phosphorylation	Involved in synthesis of ribosomal subunits	Contains euchromatin which is transcriptionally active

2. The diagram below shows a section of a cell surface membrane.



Which of the following statements are correct?

I Structure **A** is found only on the extracellular surface of the membrane.

II Structure **B** may contain a channel that is hydrophobic to allow ions to move across the membrane.

III Structure **B** is able to bind to certain substances and change its conformation to transport these substances into the cell.

IV Structure **C** allows transport of substances across the membrane only in the presence of ATP.

V Structure **D** is found only on one surface of the membrane.

A **I, II, III** and **IV**

B **II, III** and **V**

C **I** and **III** only

D **I** and **V** only

3. A gland cell capable of producing large quantities of sex hormone estrogen would be likely to have an abundance of

A lysosome

B ribosomes

C rough endoplasmic reticulum

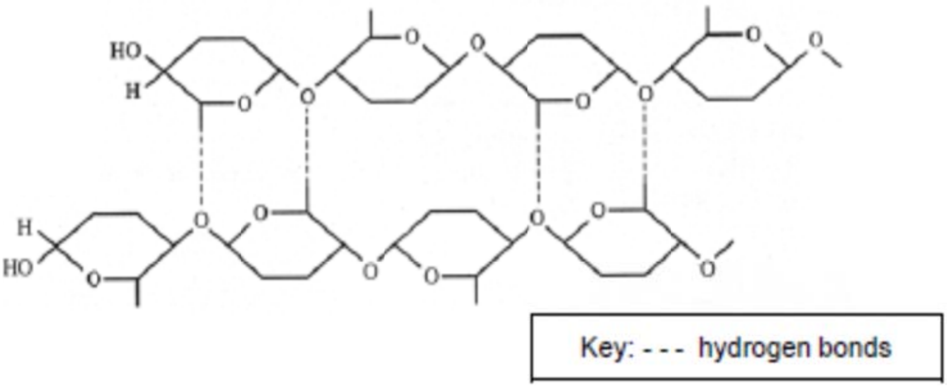
D smooth endoplasmic reticulum

4. The cell membrane of an African desert rat and that of the Arctic red fox are compared to investigate their composition. Which of the following is the most probable makeup of the respective membranes?

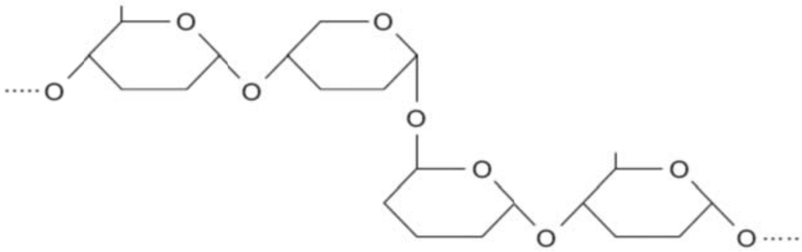
	Amount of saturated fatty acids		**Amount of cholesterol**	
	Arctic red fox	**Desert rat**	**Arctic red fox**	**Desert rat**
A	Low	High	Low	High
B	High	Low	Low	Low
C	Low	High	High	High
D	High	Low	High	Low

5. The figure below shows part of the molecular structure of two polymers, **A** and **B**.

Polymer A:



Polymer B:



Which of the following statement are true?

- **I** **A** contains β-glucose while **B** contains α-glucose.
- **II** **A** is an important food source for animals which cannot be digested by humans, due to presence of strong glycosidic bonds.
- **III** **A** may associate in groups to form microfibrils, which are arranged in larger bundles to form macrofibrils.
- **IV** **B** is a storage molecule which is held by many inter-chain hydrogen bonds.

A **I, II** and **III**

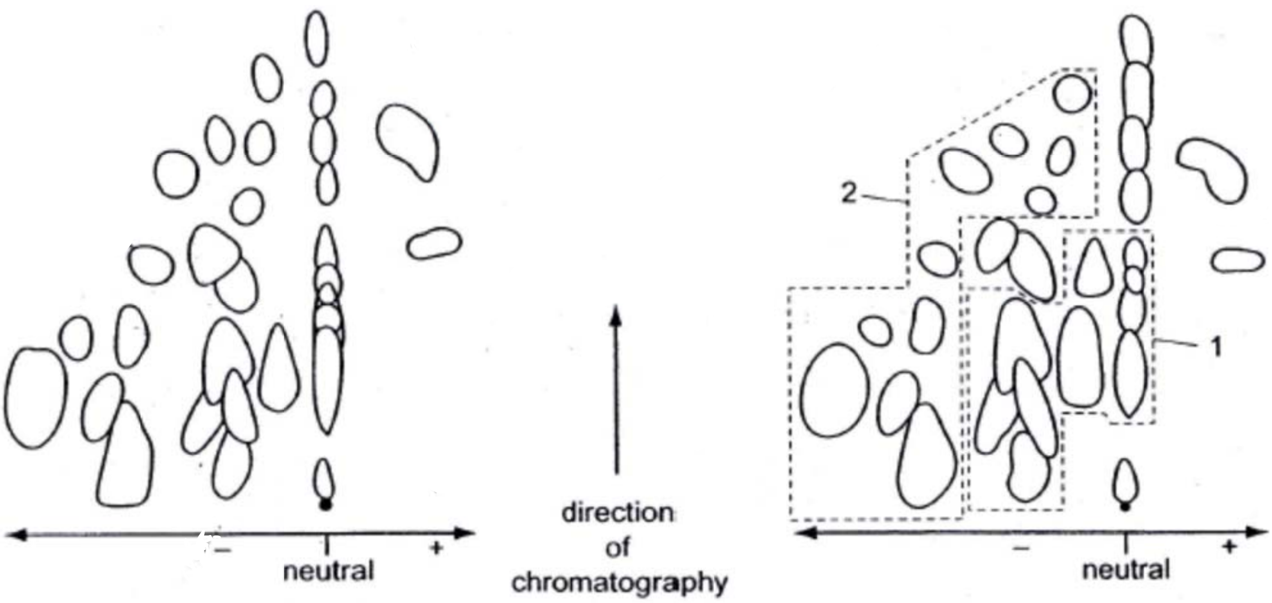
B **II** and **III**

C **I** and **III**

D All of the above

6. Early studies on sickle cell haemoglobin involved investigating the results of changed DNA nucleotide sequence. Haemoglobin was treated with a protease that hydrolysed it into several fragments. These were then separated by their charge using electrophoresis on a paper support. The paper was turned through 90^0 and chromatography used to separate the mixture. The more soluble the amino acid, the further it is carried by the solvent as it moves in the direction of chromatography.

The diagrams show the results for normal and sickle cell haemoglobin.

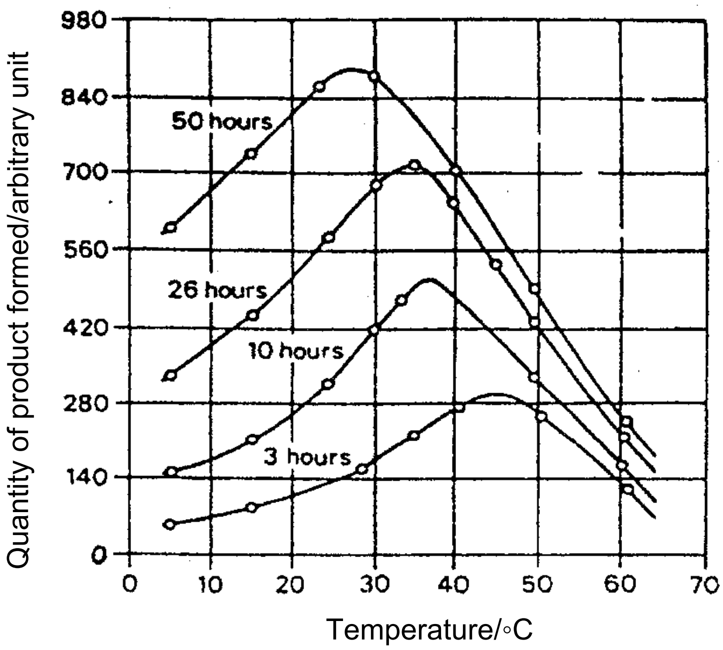


normal haemoglobin　　　　sickle cell haemoglobin

Which describes the changes to haemoglobin and the fragments resulting from the changed DNA nucleotide sequence?

	Region of the diagram that shows the fragment containing the changed amino acid	Effect of changed DNA nucleotide sequence on amino acid sequence and charge of fragment
A	1	Glutamate (hydrophilic and polar) changed to valine (hydrophobic and non-polar), no longer balancing the charge of the rest of the fragment
B	1	Valine (hydrophobic and non-polar) changed to glutamate (hydrophilic and polar), more nearly balancing the charge of the rest of the fragment
C	2	Glutamate (hydrophilic and polar) changed to valine (hydrophobic and non-polar), no longer balancing the charge of the rest of the fragment
D	2	Valine (hydrophobic and non-polar) changed to glutamate (hydrophilic and polar), more nearly balancing the charge of the rest of the fragment

7. The graph below shows the quantity of the product formed when samples containing the same concentration of enzyme and substrate were kept at different temperature for four different durations.



Which statement best explain why the optimum temperature is lowered if the duration of incubation is increased?

A There is an increase in the denaturation of enzymes at high temperature.

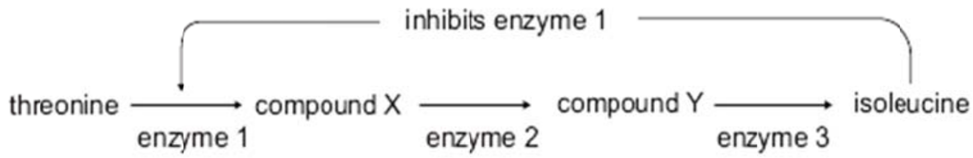
B The activation energy of the reaction is lowered at high temperature.

C The product formed acts as an allosteric activator to enhance enzyme activity.

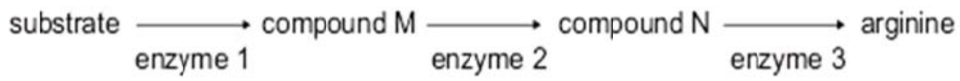
D Cooperativity between subunits of the enzyme changes shape of active sites to enhance substrate binding.

8. In the production of isoleucine from threonine (**pathway P**), the end product acts as an inhibitor of the first enzyme in the pathway. In the production of argiinine (**pathway Q**), the end product has no influence on other enzymes in the pathway. The two pathways are shown in the figure.

Pathway P:



Pathway Q:



Which of the following conclusions is correct?

I	In **pathway P**, if the production of enzyme 3 stops, there would be increased production of isoleucine.
II	In **pathway Q**, if the production of enzyme 3 stops, there would be decreased production of arginine.
III	In **pathway P**, provided all enzymes are present, the production of isoleucine would be continuous if there was a continuous supply of threonine.
IV	In **pathway Q**, provided all enzymes are present, the production of arginine would be continuous if there was a continuous supply of substrate.

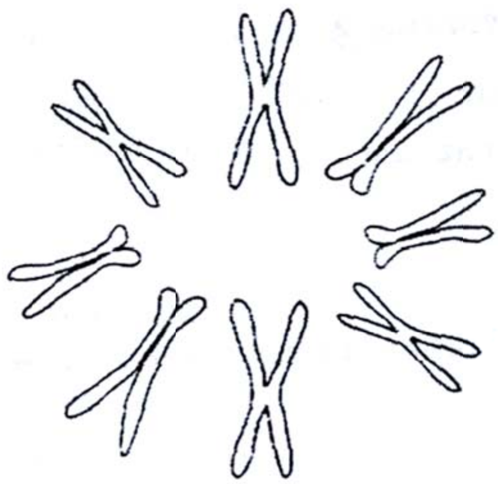
A **I** and **III** only

B **I** and **IV** only

C **II** and **III** only

D **II** and **IV** only

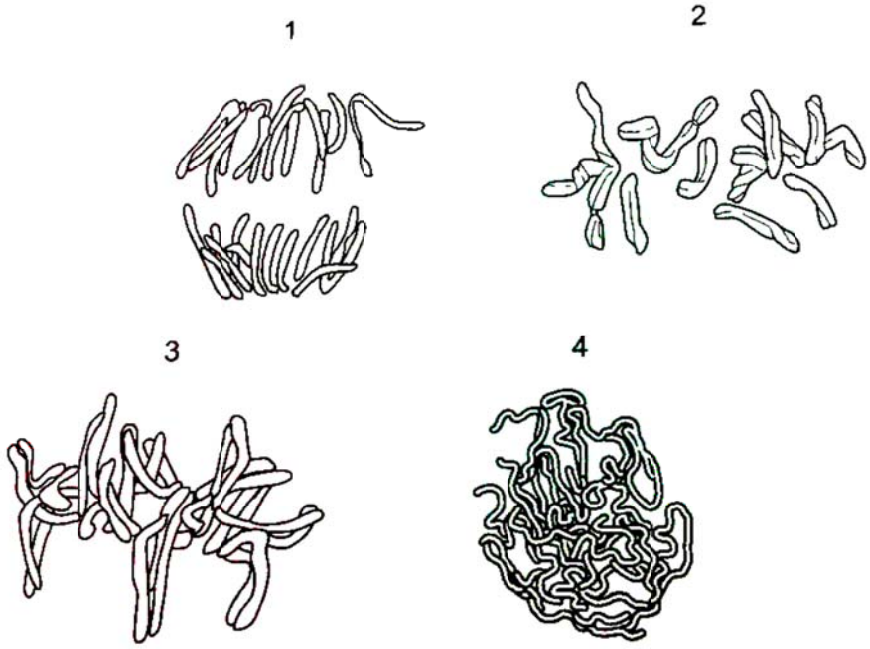
9. The diagram shows the behaviour of chromosomes in an animal cell during a particular stage of cell division.



Which of the following correctly describes the chromosomes shown?

	Stage of cell division	Number of chromosomes in a haploid set	Total number of DNA molecules
A	Metaphase	4	16
B	Prophase	4	16
C	Prophase	8	32
D	Metaphase	8	32

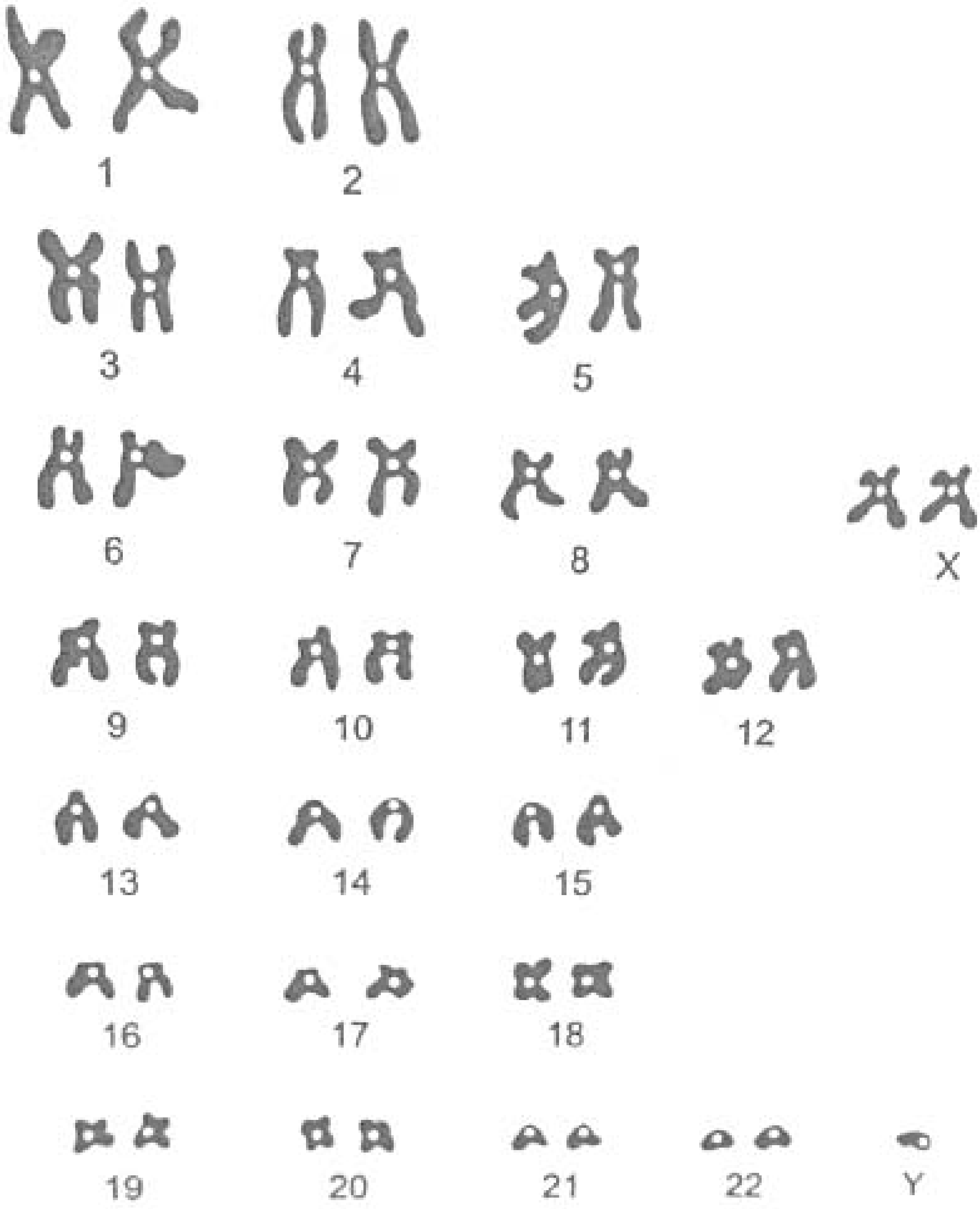
10. The drawing show stages of the mitotic cell cycle.



What would be correct?

	Stage of mitosis	Name of the stages	Description
A	1	Anaphase	Non-kinetochore microtubule lengthens, pulling chromosomes to opposite poles.
B	2	Prophase	The chromosomes move towards the equator to line up end to end of each other.
C	3	Metaphase	The homologous chromosomes pairs up at the equator.
D	4	Prophase	DNA replicates, each will form 2 DNA daughter molecules which condensed to form 2 identical sister chromatids.

11. The figure below shows a human karyotype.



What can be concluded from the karyotype provided?

A There was non-disjunction during meiosis I in the mother.

B There was non-disjunction during meiosis II in the father.

C One contributory gamete to the zygote is an egg with an X and a Y chromosome.

D One contributory gamete to the zygote is a sperm containing two X chromosomes.

12. Two genes involved in coat color of goats are at loci on different chromosomes.

The color gene C causes the hairs to be uniform color and has three alleles.

- C^{DB} giving dark brown hairs
- C^{B} giving black hairs
- C^{MB} giving medium brown hairs

A dominant allele of the agouti gene (A^G) causes the development of white hairs between the colored hairs giving the coat a shaded appearance.

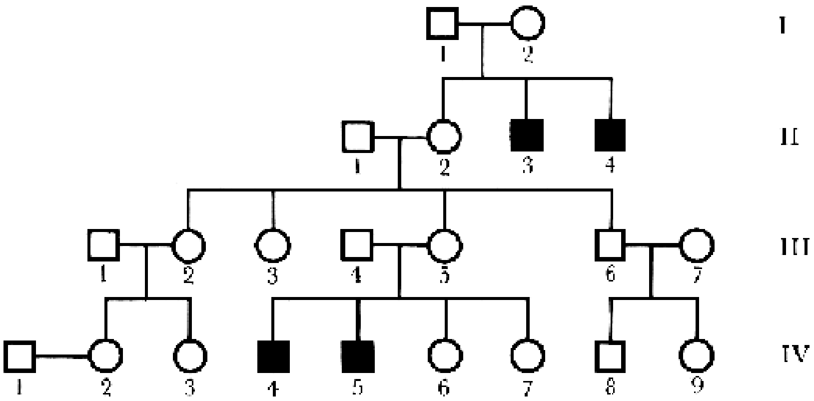
The table shows the results of crosses between a male goat and two female goats.

Cross	**Parents**	**Offspring**
1	Black agouti male **x** Uniformly dark brown female	50% uniform, 50% agouti; 50% dark brown, 50% black
2	Black agouti male **x** Medium brown agouti female	all agouti; 50% black, 50% medium brown

Which row shows the possible genotypes of the male and female goats?

	Black agouti male	Uniformly dark brown female	Medium brown agouti female
A	$C^BC^{MB}A^GA^G$	$C^BC^{DB}A^GA^g$	$C^{MB}C^{MB}A^GA^G$
B	$C^BC^{MB}A^GA^g$	$C^BC^{DB}A^gA^g$	$C^BC^{MB}A^GA^G$
C	$C^BC^{DB}A^GA^G$	$C^{DB}C^{MB}A^GA^g$	$C^BC^{DB}A^GA^g$
D	$C^BC^{MB}A^GA^g$	$C^BC^{DB}A^gA^g$	$C^{MB}C^{MB}A^GA^G$

13. Haemophilia is a rare genetic disorder in which the blood does not clot normally. The mutation that causes haemophilia is located on the X-chromosome. The pedigree below shows the inheritance of haemophilia in a family. Squares represent males while circles represent females. Dark symbols represent affected individuals.



What is the probability that a son born to IV-2 will be a haemophiliac?

A 0.5

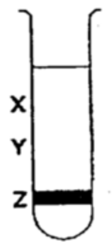
B 0.25

C 0.125

D 0.0625

14. A culture of bacteria had all its DNA labelled with the heavy isotope of nitrogen, ^{15}N. The bacterial culture was then allowed to reproduce using nucleotides containing normal ^{14}N. The DNA was examined using a centrifuge after one generation and again after two generations.

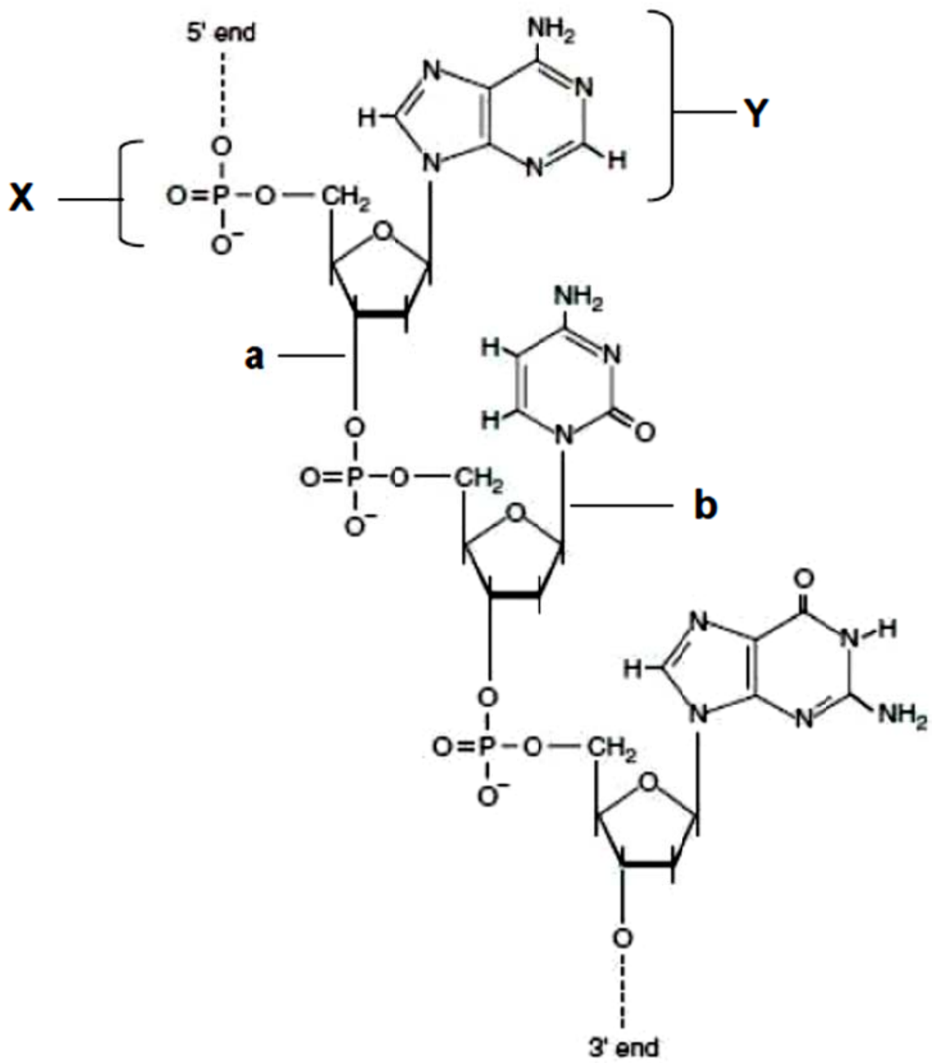
The diagram shows the position of the DNA band at **Z** in the centrifuge tube when the DNA was first labelled.



Where would the DNA be found after the second and after the third cell generations?

	After second generation	After third generation
A	Half at X and half at Y	One quarter at X and three quarter at Y
B	Half at X and half at Y	One quarter at Y and three quarter at X
C	One quarter at X and three quarter at Y	Half at X and half at Y
D	One quarter at Y and three quarter at X	Half at X and half at Y

15. The figure below shows the components of a biological molecule.



Which of the following statements are true about this molecule?

I X enables this molecule to move from the negative electrode to the positive electrode during gel electrophoresis.

II Y can be degraded at 95 °C, the highest temperature used in PCR.

III Formation of **a** involves the removal of a water molecule.

IV **a** and **b** are covalent bonds that can be hydrolysed by the same enzyme.

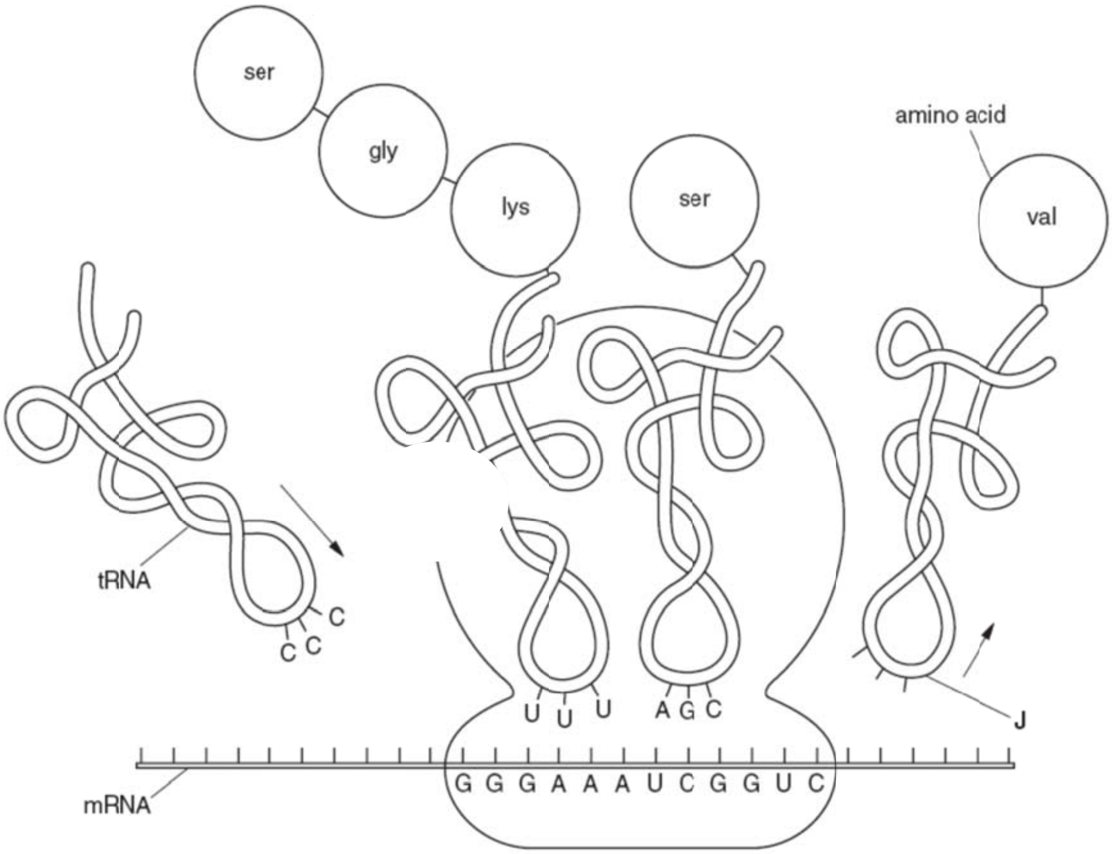
A **I** and **II**

B **I** and **III**

C **II** and **III**

D **II** and **IV**

16. The diagram illustrates the process of translation.



Which of the following statements is / are true?

I	The base sequence at structure **J** is 3'–CCA–5'.
II	The anticodon of an incoming aminoacyl-tRNA base pairs with the complementary mRNA codon in the P site of the ribosome.
III	The movement of the ribosome is from the right to the left of the diagram above.
IV	A release factor causes the addition of a water molecule instead of an amino acid to the polypeptide chain, hydrolysing the completed polypeptide from the tRNA in the A site. This brings about the release of the polypeptide through the exit tunnel of the ribosome's large subunit.

A **IV** only

B **I** and **III** only

C **I, II** and **III** only

D None of the above

17. The following shows part of a normal DNA sequence of a gene and the first 5 amino acids it codes for. A mutation occurred resulting in a different polypeptide sequence.

Normal DNA sequence : G A T C A T C A T A T G A G C

Normal polypeptide sequence : Leu – Val – Val – Tyr – Ser

Mutant polypeptide sequence : Leu – Val – Ile

The table shows 6 amino acids and their corresponding codons.

Amino Acid	*mRNA Codon*
Ile	AUA
Gly	GGA
Leu	CUA / CUC
Met	AUG
Ser	UCG
Thr	ACU
Tyr	UAC / UAU
Val	GUA / GUC
STOP	UAA / UAG / UGA

Which of the following mutations is most likely to have occurred?

A Deletion at base 7
B Deletion at base 7 and 8
C Inversion from base 6 to 12
D Substitution at base 7

18. One hereditary disease in humans, called xeroderma pigmentosum (XP), makes homozygous individuals exceptionally susceptible to UV-induced mutation damage in the cells of exposed tissue, especially skin. Without extraordinary avoidance of sunlight exposure, patients soon succumb to numerous skin cancers. Which of the following best describes this phenomenon?

A inherited cancer taking a few years to be expressed
B embryonic or fetal cancer
C inherited predisposition to mutation
D inherited inability to repair UV-induced mutation

19. Which one of the following substances, when added, would directly result in a decline in ATP production in glycolysis?

I A chemical that would bind to NAD^+ irreversibly and induces its reduction to NADH.

II An inhibitor that has a similar structure to glucose but cannot be broken down by respiratory enzymes.

III A chemical that creates an anaerobic environment by combusting in oxygen

IV A reagent that binds to the active site of ATPase permanently

A **I** and **II** only

B **III** and **IV** only

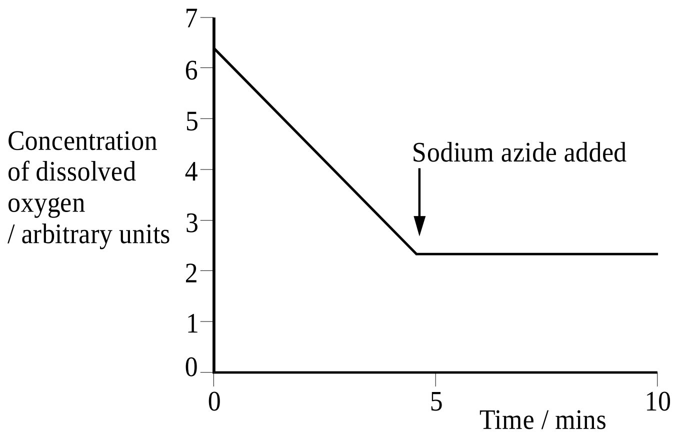
C **I**, **II** and **IV** only

D All of the above

20. Active mitochondria can be isolated from liver cells. If these mitochondria are then incubated in a buffer solution containing a substrate, such as succinate, dissolved oxygen will be used by the mitochondria. The concentration of dissolved oxygen in the buffer solution can be measured using an electrode.

An experiment was carried out in which a suspension of active mitochondria was incubated in a buffer solution containing succinate, an intermediate of the Krebs cycle. The concentration of dissolved oxygen was measured every minute for five minutes. A solution containing sodium azide was then added to this preparation and the concentration of dissolved oxygen was measured for a further five minutes.

The results are shown in the graph below.



Which of the following statements does not explain the results shown in the graph?

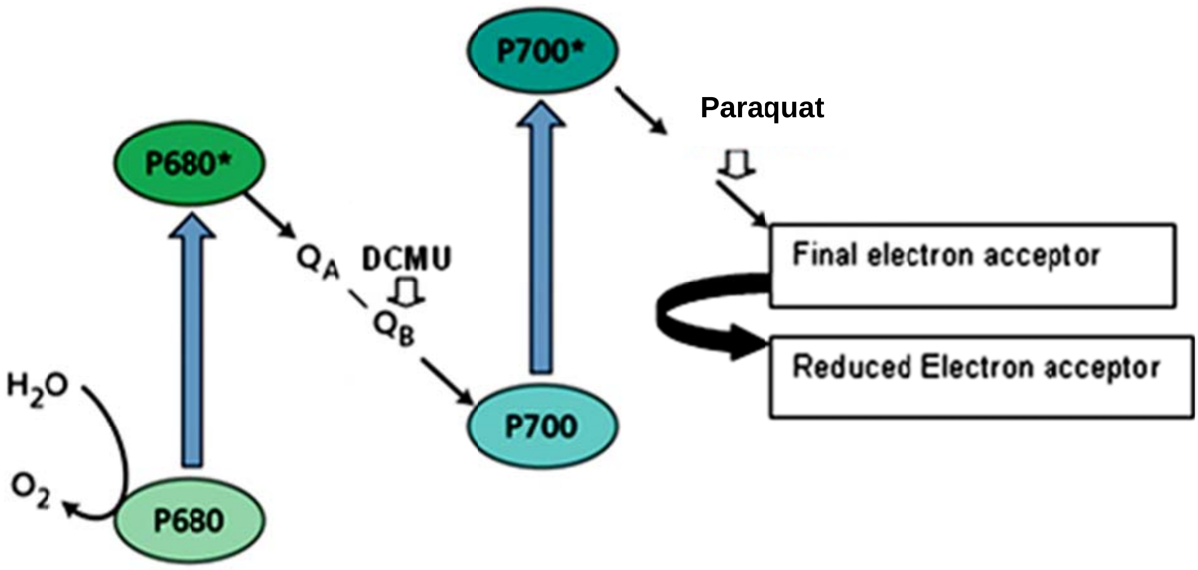
A The decrease in the concentration of dissolved oxygen in the first 5 minutes is due to oxygen combining with hydrogen ions and electrons to form water.

B **Sodium azide inhibits cytochrome oxidase from binding to water.**

C When sodium azide is added, the electron flow down the electron transport chain is inhibited.

D Succinate cannot be oxidised when sodium azide is added.

21. The diagram shows the electron transport chain embedded in the thylakoid membrane.

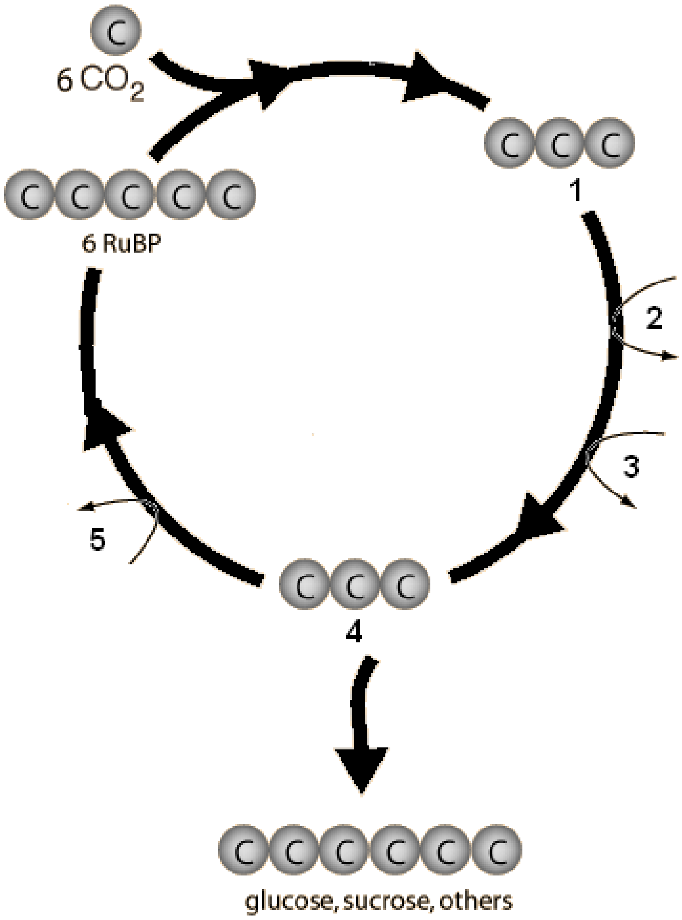


DCMU and paraquat are poisons that disrupt electron transport at the respective positions indicated in the diagram. These two poisons are added separately to isolated chloroplasts. Which of the following ***CORRECTLY*** represents the outcomes in the presence of DCMU and paraquat?

(The sign “+” means produced and the sign “-“ means not produced.)

	DCMU			Paraquat		
	NADPH	**Oxygen**	**ATP**	**NADPH**	**Oxygen**	**ATP**
A	+	-	+	-	-	+
B	-	+	-	-	+	+
C	-	+	+	+	+	-
D	-	-	-	-	-	+

22. The Calvin cycle involves several reactions.



Which row correctly identifies the molecules or reactions at 1 to 5?

	1	2	3	4	5
A	glycerate phosphate	ATP→ADP	NADPH→NADP	triose phosphate	ATP→ADP
B	glycerate phosphate	ATP→ADP	NADH→NAD	triose phosphate	ATP→ADP
C	glycerate phosphate	ADP→ATP	NADP→NADPH	triose phosphate	ADP→ATP
D	triose phosphate	ATP→ADP	NADPH→NADP	glycerate phosphate	ATP→ADP

23. A single clone of E. coli was cultured at 37°C for 2000 generations. A single clone was then extracted from this population and divided into replicates that were then cultured at either 32 °C, 37 °C, or 42 °C for a total of another 2000 generations. Adaptation of the new lines to 32 °C and 42 °C were periodically measured by comparing these selection lines against the ancestor population.

By the end of the experiment, the lines cultured at 32 °C were shown to be 10% fitter that the ancestor population (at 32 °C), and the line cultured at 42°C was shown to be 20% more fit than the ancestor population (at 42 °C). The replicate line that was cultured at 37°C showed little improvement over the ancestral line.

From the information, which of the following is the *best* conclusion?

A There are three species of E. coli after 4000 generations.

B In the ancestral line, there exist only 3 variations of E. coli, i.e. E. coli with tolerance to 32 °C, 37 °C, or 42 °C.

C The selection agent selects for the genes that enable E. coli to better survive in differing temperatures.

D The selection agent selects for phenotypic traits that enable E. coli to show adaptations to high and low temperatures.

24. The table shows the functions of three bones, M, I and S, in three classes of vertebrates.

Each bone arises in the same way in embryos of the three classes.

Class	**M**	**I**	**S**
Fish	part of the articulation of upper and lower jaws	part of the articulation of upper and lower jaws	attaches the upper and lower jaws to the skull
Amphibian	part of the articulation of the lower jaw	part of the articulation of the lower jaw	transmits sound to the inner ear
Mammals	transmits sound to the inner ear	transmits sound to the inner ear	transmits sound to the inner ear

How do the functions of these bones support Darwin's theory of natural selection?

A They show convergent evolution of analogous structures.

B They show divergent evolution of analogous structures.

C They show how analogous structures have become adapted to perform different functions.

D They show how homologous structures have become adapted to perform different functions.

25. Which of the following characteristics is undesirable in cloning vectors used in genetic engineering?

E high copy number

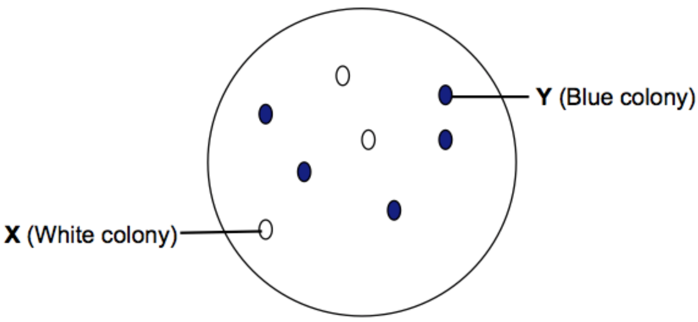
F vulnerable at several sites to a restriction enzyme

G control their own replication

H small in size

26. pGEM-T is a 7.25kb plasmid cloning vector often used in creating recombinant DNA. It contains two selectable marker genes, TetR and lacZ. The multiple cloning site (MCS) of the cloning vector is located within the lacZ gene.

Using SacII as the restriction enzyme, a 2.31kb insert (gene of interest) is cloned into the MCS of plasmid pGEM-T and the recombinant plasmid is transformed into E. coli cells. The transformants are then plated on medium containing tetracycline and X-Gal. The following result is obtained.



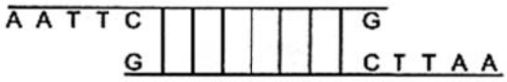
Colonies **X** and **Y** are isolated and the plasmid DNA from the colonies is extracted. The extracted plasmids are then digested with *Sac*II. Which of the following correctly shows the fragments obtained?

	Colony X	**Colony Y**
A	A 9.56kb fragment	A 7.25kb fragment
B	A 7.25kb fragment	A 9.56kb fragment
C	7.21kb and 0.34kb fragments	7.21kb and 2.31kb fragments
D	7.21kb and 0.34kb fragments	7.21kb and 0.34kb fragments

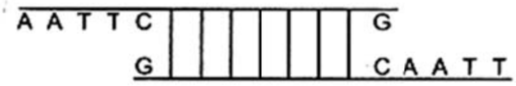
27. In genetic engineering, a restriction enzyme is used to cut plasmid DNA at a specific target site. The enzyme recognises a sequence of six bases and forms sticky ends.

Which diagram of such a cut section of DNA is correct? **A**

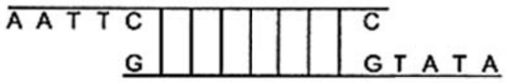
A



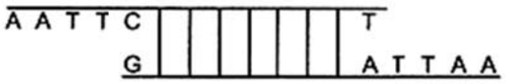
B



C



D



28. Which uses of the information from the human genome project are generally considered to be unethical?

1 an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases

2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships

3 cytologists developing tests for only some defective genes

4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes

5 genetic councillors giving specific lifestyle information only to people genetically predisposed to risks

6 parents choosing embryos for implantation only after ante-natal tests for acceptable genes

A 1 and 3

B 1 and 6

C 2 and 6

D 3 and 4

29. Which of the following is not true of adult stem cells during tissue repair?

A The stem cells must have active telomerase.

B The different checkpoints in the cell cycle of the stem cells are activated.

C Mitosis of the stem cells is induced without any stimulus.

D The stem cells will stop dividing after the damaged cells are replaced.

30. In terms of containment (prevention of genetic pollution of wild plant or non-GM crop populations), which of the following is an advantage of chloroplast transformation (introducing foreign genes into chloroplasts of host plants) over nuclear transformation?

A Chloroplasts are surrounded by a double membrane.

B There are no chloroplasts in pollen of most plant species.

C Chloroplasts are smaller than the nucleus.

D There is no DNA in chloroplasts.

END OF PAPER